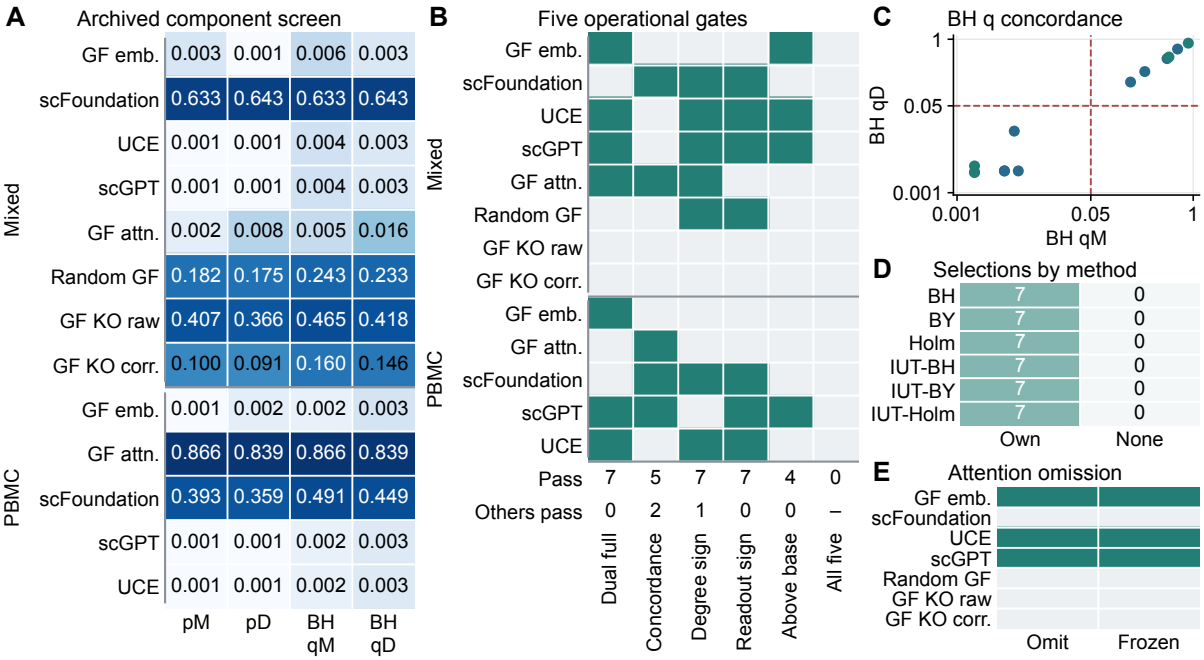
A
Archived component screen
Mixed
PBMC
GF emb. 0.003 0.001 0.006 0.003
scFoundation 0.633 0.643 0.633 0.643
UCE 0.001 0.001 0.004 0.003
scGPT 0.001 0.001 0.004 0.003
GF attn. 0.002 0.008 0.005 0.016
Random GF 0.182 0.175 0.243 0.233
GF KO raw 0.407 0.366 0.465 0.418
GF KO corr. 0.100 0.091 0.160 0.146
GF emb. 0.001 0.002 0.002 0.003
GF attn. 0.866 0.839 0.866 0.839
scFoundation 0.393 0.359 0.491 0.449
scGPT 0.001 0.001 0.002 0.003
UCE 0.001 0.001 0.002 0.003
pM
pD
BH qM
BH qD
B
Five operational gates
Mixed
PBMC
GF emb.
scFoundation
UCE
scGPT
GF attn.
Random GF
GF KO raw
GF KO corr.
GF emb.
GF attn.
scFoundation
scGPT
UCE
Pass 7 5 7 7 4 0
Others pass 0 2 1 0 0 –
Dual full
Concordance
Degree sign
Readout sign
Above base
All five
C
BH q concordance
BH qD
BH qM
1
0.05
0.001
0.001
0.05
1
D
Selections by method
BH 7 0
BY 7 0
Holm 7 0
IUT-BH 7 0
IUT-BY 7 0
IUT-Holm 7 0
Own
None
E
Attention omission
GF emb.
scFoundation
UCE
scGPT
Random GF
GF KO raw
GF KO corr.
Omit
Frozen
F
Readout signs
Mixed
PBMC
Embedding
Attention
−0.004
0
0.004
Own-degree ρ
G
Row-shuffle reference
GF emb.
scFoundation
UCE
Co-expr.
−0.002
0.002
0.006
Observed vs null ρ
H
Attention direction
Sym. −0.027
TF→target −0.043
Target→TF −0.009
Sym. | cx −0.035
TF→tgt | cx −0.048
−0.04
0
ρ (named design)
I
Knockout control
Marginal
raw 0.069
corr. 0.033
| co-expr.
raw 0.071
corr. 0.035
0.00
0.04
0.08
Historical ρ